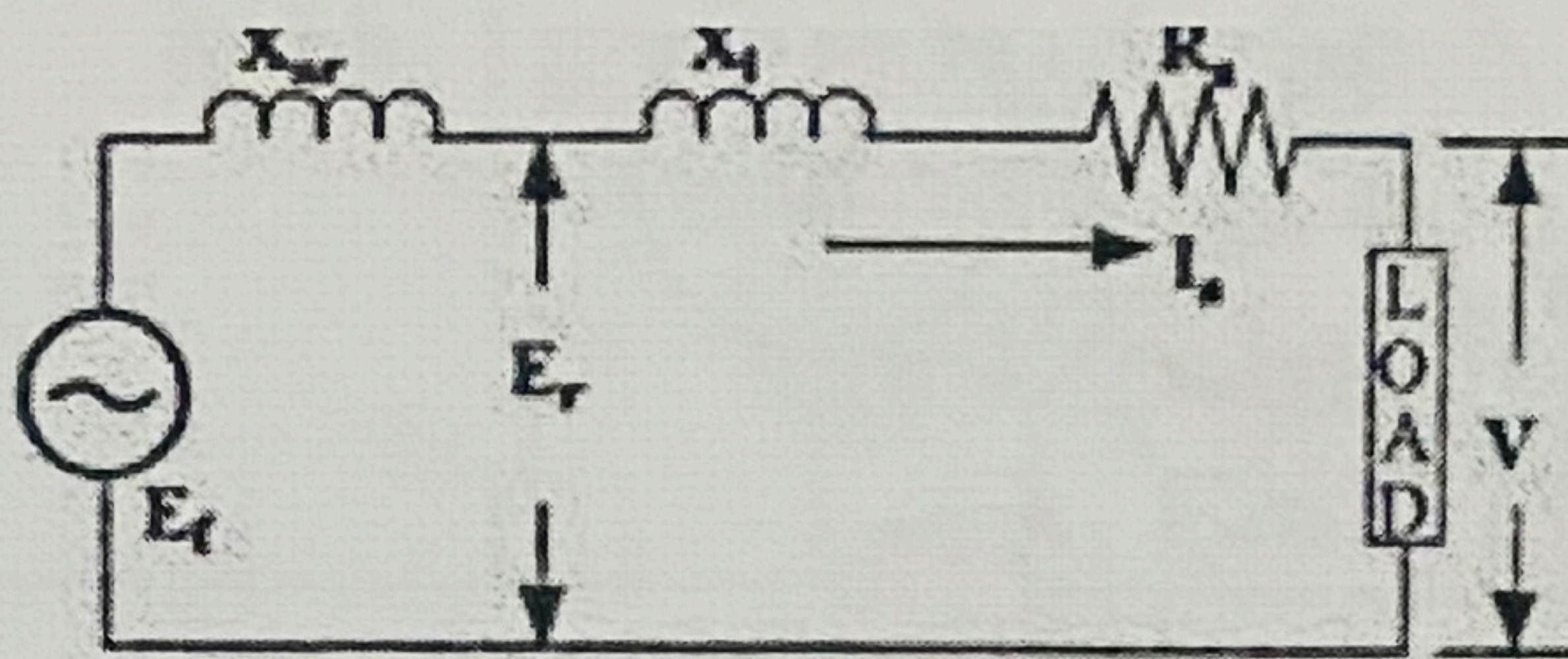




Circuit characteristic (OCC) : The open circuit characteristic of an alternator is a curve of the armature terminal voltage on open circuit as a function of field excitation when the machine is running at synchronous speed.



Short circuit characteristic (SCC): It is the plot of short circuit armature current as a function of field current when the machine is running at synchronous speed.

Zero power factor curve (ZPFC): Zero power factor characteristic of an alternator gives the variation of terminal voltage with field current, when the alternator is delivering its full load current to a zero power factor (lagging) load.

PROCEDURE

a. Open Circuit & Short Circuit Characteristics (OCC & SCC)

Make the connections as shown in diagram.

Precautions - Initial settings:

- TPST in open position
- DPST1 and DPST2 in open position
- Motor field rheostat in minimum position
- Potential divider in minimum voltage position

Switch on the DC supply to the DC motor by closing the switch DPST₁. Start the DC shunt motor using 3-point starter. Increase the resistance of DC motor field rheostat and drive the alternator at rated speed. Now, dc supply is given to the alternator field winding and for different values of field current, note down the open circuit voltage across the armature terminals. Take care to keep the speed constant (rated value) throughout the experiment. The above procedure is repeated until the open circuit voltage reaches 120% of rated value. Open circuit voltage/phase E_f vs. field current I_f gives OCC.

For SCC, reduce the armature voltage to zero by bringing the potential divider to minimum voltage position. Now, close the TPST switch. By varying the potential divider, increase the current through the short circuited armature up to rated value. Note both the ammeter readings. I_{sc} vs. I_f gives SCC.